

Oracle EPM Predictive Cash Forecasting

Machine Learning Forecasting - Advanced Lab Guide

Product Design Documentation • Implementation Guide • Training Material • Technical Reference • Troubleshooting Handbook

Important product-status note

Oracle documentation describes Machine Learning forecasting methods, including short-term and mid-term approaches, but current Predictive Cash Forecasting administration/overview documentation still states that Machine Learning support will be available in a future update. This guide therefore serves as advanced design, readiness, training, and implementation-preparation material; validate feature availability in your target Oracle release before production deployment.

Guide Property	Design
Primary use case	Receivables cash-inflow forecasting using customer payment behavior
Focus	Short-term invoice-level ML plus mid-term historical ML concepts
Training depth	Advanced architecture, eligibility, data preparation, accuracy, scenarios, controls
Audience	Oracle EPM consultants, treasury teams, finance architects, administrators
Branding	One BISP logo per page; professional Oracle-inspired visual language

BISP Trainings

1. Purpose and Learning Outcomes

Learning objective

Advanced students should understand when ML adds value over Smart Drivers, Statistical Forecasting, and Driver-Based Forecasting; how to prepare training data; how forecast accuracy changes by horizon; and how ML outputs should be governed within the cash-forecast process.

- Explain short-term vs mid-term ML forecasting concepts in PCF.
- Assess billing models and transaction lifecycles for ML eligibility.
- Prepare a high-quality training set with appropriate exclusions.
- Understand Oracle guidance on weekly retraining and forecast-uptake frequency.
- Compare ML accuracy with Average Days Delinquent or other baseline methods.
- Design exception handling for new customers, migrated transactions, credit memos, and atypical records.
- Blend ML with other forecast methods by line item, entity, and horizon.
- Build a production-ready monitoring, governance, and troubleshooting model.

2. Current Oracle ML Context and Availability

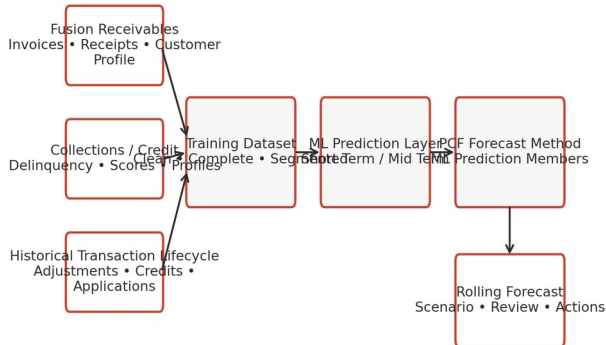
Oracle documentation point	Implementation interpretation
PCF documentation defines Short Term ML Prediction and Mid Term ML Prediction members	The application design anticipates both invoice-level and historical ML approaches.
Short-term ML is primarily suited to daily/short-term invoice-level prediction	Use it where transaction detail and customer payment behavior are strongest.
Mid-term ML is based on historical data	Use historical behavioral patterns for broader short/mid-term projections.
Current PCF overview/admin content still labels ML as a future update	Do not assume production GA in every environment; validate the target release.
ERP readiness material describes ML alongside other PCF methods	Prepare architecture, data and controls now so enablement can be accelerated when available.

Design stance

This guide does not fabricate a live ML screen. Configuration mockups illustrate product-design intent and implementation-readiness concepts based on Oracle-documented forecast-method members and guidance.

3. ML Forecasting Reference Architecture

Machine Learning Forecasting Reference Architecture

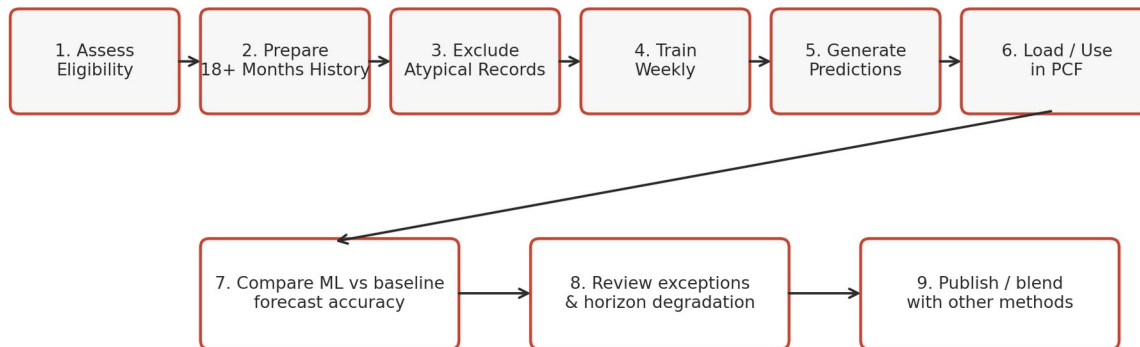


Implementation note: Oracle documentation describes ML forecast-method design, while current PCF administration content still flags ML support as a future update; validate release availability before production deployment.

The target architecture is centered on Oracle Fusion Receivables because ML accuracy depends on invoice lifecycle data, customer payment behavior, credit attributes, collections profiles, receipt applications, adjustments, and other transactional signals. The training dataset should preserve accurate business attributes while excluding records that do not represent genuine payment behavior.

4. ML Forecasting Operating Process

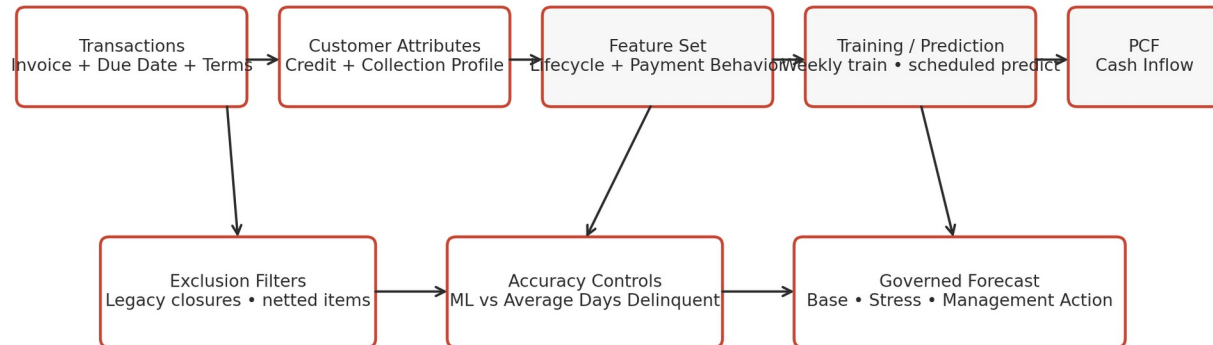
ML Forecasting End-to-End Process



1. Assess whether the business captures due-date-based receivables and customer lifecycle data in Fusion Receivables.
2. Prepare a minimum of roughly 18 months of historical transactions with complete lifecycle activities.
3. Define and apply exclusions for atypical or misleading records.
4. Train or refresh the model on the solution cadence; Oracle guidance describes weekly training.
5. Generate predictions at a cadence aligned to the organization cash-forecast frequency and transaction volatility.
6. Use or load the ML forecast into PCF for the relevant Receivables cash-inflow line items.
7. Compare ML forecast accuracy to a benchmark such as Average Days Delinquent.
8. Review exceptions and deterioration in accuracy as the horizon extends beyond standard payment terms.
9. Publish, override, or blend the forecast through controlled versions and governance.

5. Data Flow, Features, and Training Lineage

ML Data Flow and Training Lineage



Feature / data family	Examples	Why it matters
Invoice lifecycle	Transaction date, due date, payment terms, amount, line details	Core behavioral inputs for predicting payment timing.
Receipt behavior	Receipt applications, lateness, partial payment patterns	Shows how customers actually settle obligations.
Credit profile	Credit class, scores, limits, case-folder context	Adds information about customer credit quality.
Collections profile	Collector, delinquency, strategy, dunning activity	Adds behavioral information about collection risk.
Customer master	Segmentation, profile, account attributes	Helps model customer-specific payment behavior.
Exclusion attributes	Source/type markers for atypical records	Prevents contamination of training data.

6. Business-Model Eligibility Framework

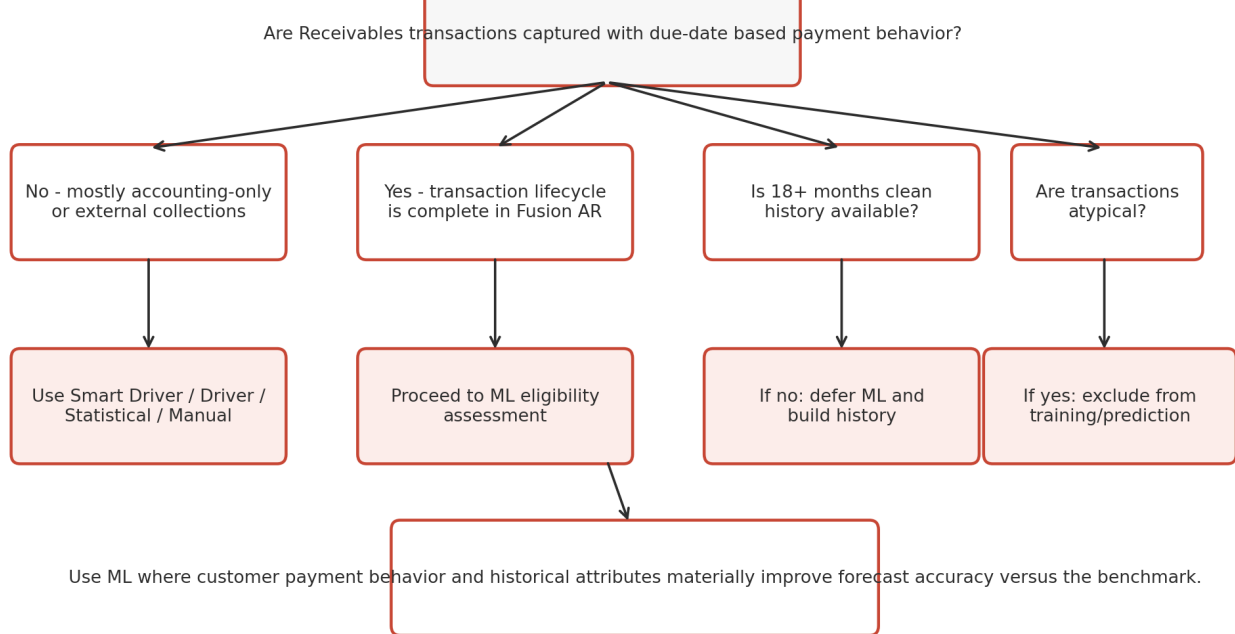
Billing model	Description	ML suitability
Model 1	Rating/charges external; billing, collections and cash application in Fusion Financials	Strong fit because transaction lifecycle is available in Fusion.
Model 2	Billing is replicated into Fusion for collections/cash application	Potential fit; exclude immediate-pay populations and ensure replicated attributes are accurate.
Model 3	Billing and collections occur outside Fusion; only accounting data enters ERP	Lower suitability because customer payment lifecycle is incomplete in Fusion.

Consultant takeaway

ML readiness is a business-process question before it is a technical question. If the lifecycle that explains customer payment behavior is not present in Fusion, the model has much less to learn from.

7. Decision Tree - Is ML the Right Method?

Decision Tree - Is ML the Right Forecast Method?



- If transaction-level payment behavior is incomplete, prefer Smart Drivers, Statistical Forecasting, Driver-Based, Trend-Based, or Manual methods.
- If clean history is insufficient, first build a reliable data history rather than forcing model deployment.
- If a subset of transactions is atypical, exclude it through Receivables transaction sources/types or other supported segmentation.
- Use ML only where it demonstrates measurable forecast accuracy improvement over the baseline.

8. Training Data Requirements

Requirement	Recommended design
History depth	At least 18 months of historical transactions with complete lifecycle information.
Business similarity	Historical transactions should resemble current transactions and current operating model.
Lifecycle completeness	Receipt applications, adjustments, credits and closures should be accurately recorded.
Customer attributes	Credit and collections data should reflect how billing and payments actually operate.
Exclusion policy	Remove converted/legacy closures and atypical netting/non-settlement records.
Data-quality ownership	Assign business and technical owners for feature quality, not just pipeline success.

9. Exclusion and Data-Cleansing Rules

Record type	Why exclude / treat carefully
Immediately-paid credit card / wallet / bank-transfer transactions	May distort due-date-based payment behavior when mixed with true credit transactions.
Legacy-converted open transactions closed immediately in Fusion	Do not represent authentic payment behavior in the current operating model.
Negative invoices / positive credit memos / zero-dollar netted invoices	Atypical mechanics can mislead the learning process.
Intercompany receivables without actual settlement	Can create artificial patterns unrelated to external cash collection.
New geography / entity with different process	Historical behavior may not correlate with new operating conditions.

10. Short-Term vs Mid-Term ML Forecasting

Dimension	Short Term ML Prediction	Mid Term ML Prediction
Primary level	Invoice-level	Historical pattern / aggregate context
Best use	Daily and short-term cash forecasting	Short- to mid-term cash planning
Key dependency	Current eligible receivable transactions and customer payment behavior	Historical data continuity
Accuracy profile	Strongest near expected payment-term window	Depends on historical stability and available explanatory factors
Forecast Method member	OCF_Short Term Machine Learning Prediction	OCF_Machine Learning Prediction

11. Horizon Design and Accuracy Decay

Oracle design guidance

ML forecasting is best aligned to the dominant payment-term horizon. For example, if most customers use Net 45 terms, accuracy for forecasts through about 45 days is expected to be stronger than forecasts extending to around 90 days, because fewer currently-known transactions and explanatory factors exist farther into the future.

Horizon	Recommended treatment
D+1 to D+15	High-frequency uptake for volatile businesses; prioritize transaction-level ML.
D+16 to D+45	Core ML horizon for a Net-45 population.
D+46 to D+90	Use caution; monitor degradation and consider blending with Statistical/Driver methods.
Beyond D+90	Often better handled with broader driver, trend, statistical, or manual methods depending on line item.

12. Oracle-Inspired Configuration Mockup

Oracle-Inspired ML Forecast Method Configuration Mockup

Home > Application > Forecast Methods > Receivables Cash Inflow > Machine Learning

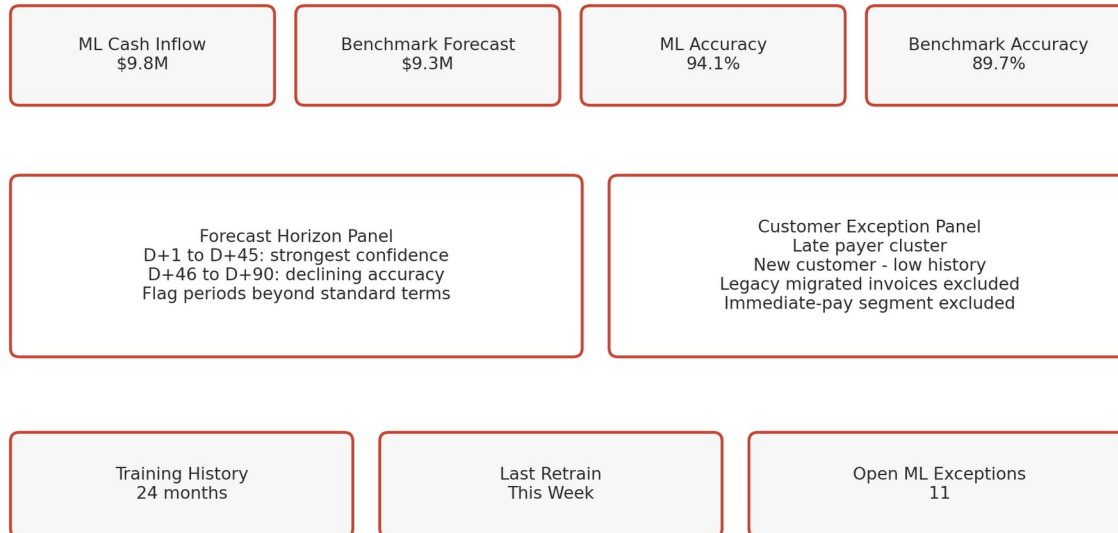
Entity US01	Method Short Term ML Prediction	Training Frequency Weekly	Exclusions Configured
Line Item Receivables Cash Inflow	Forecast Range Daily / short term	Prediction Uptake Daily / aligned to cadence	Validation Accuracy vs benchmark

Save / Validate

- ☐ Validate ML forecast-method availability in the target Oracle release.
- ☐ Confirm Receivables integration and required transactional attributes.
- ☐ Define eligible entities and line items.
- ☐ Map short-term vs mid-term forecast-method members.
- ☐ Define training cadence, prediction uptake cadence, and horizon cut-offs.
- ☐ Configure exclusions by transaction source/type or other governance criteria.
- ☐ Define benchmark method for accuracy comparison.
- ☐ Set version/scenario governance for ML output.

13. Dashboard and Monitoring Mockup

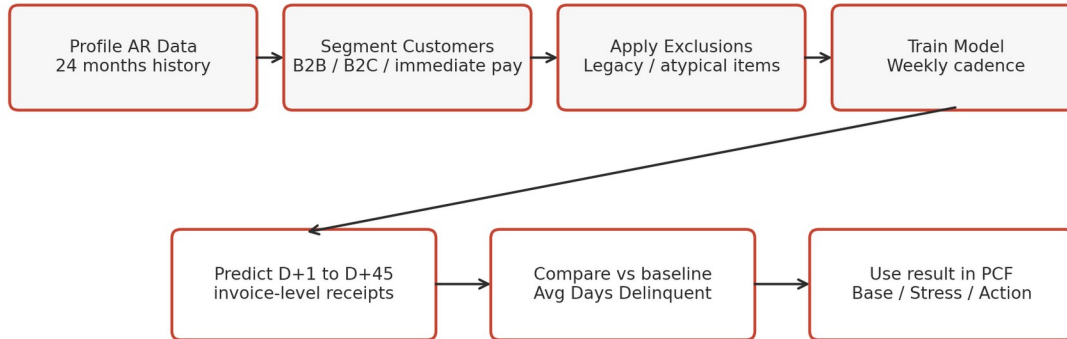
ML Forecast Dashboard Mockup



- Show ML forecast next to the benchmark, not in isolation.
- Display accuracy by forecast horizon so users see where the model weakens.
- Surface training-data coverage and last retrain time.
- Expose exception clusters such as new customers, migrated records, and excluded immediate-pay transactions.
- Use scenario analysis to show the liquidity consequence of ML uncertainty.

14. Advanced Lab Scenario

Advanced Lab Flow - ML Receivables Forecasting



Scenario

BISP Manufacturing US operates a B2B receivables model in Fusion Cloud ERP. Treasury runs a daily 60-day cash forecast. Most customers have Net 45 terms. The team has 24 months of clean receivables history and wants to determine whether ML improves short-term receipts forecasting compared with Average Days Delinquent.

Lab parameter	Value
Entity	US01
Forecast horizon	60 daily days
Dominant payment term	Net 45
Training history	24 months
Training frequency	Weekly
Prediction uptake	Daily
Benchmark	Average Days Delinquent / Smart Driver baseline
Primary KPI	Cash forecast accuracy

15. Lab Step 1 - Build the Training Population

Population	Records	Treatment
B2B due-dated invoices	18,420	Include
Immediate-pay credit card invoices	3,210	Exclude
Legacy migrated invoices closed immediately	640	Exclude
Intercompany netted receivables	510	Exclude
Zero-dollar / atypical netting items	190	Exclude
Eligible final training population	18,420	Train

Control

Record-count reconciliation must prove that every excluded population is explained and that the eligible training population is complete and reproducible.

16. Lab Step 2 - Customer Behavioral Features

Customer	Payment Term	Avg Days Late	Credit Profile	Collections Segment	Expected ML Behavior
Alpha Components	Net 30	2	Low risk	Standard	Near-due-date receipt
Beta Retail	Net 45	11	Medium risk	Watch	Later than due date
Gamma Health	Net 60	4	Low risk	Strategic	Slightly late
Delta Tech	Net 45	23	High risk	Delinquent	Material delay
Epsilon Logistics	Net 30	0	Low risk	Auto-pay	At/near due date

This simplified table illustrates why ML can outperform one common delay assumption: customer behavior is heterogeneous and correlated with payment terms, credit status, and collections treatment.

17. Lab Step 3 - Invoice-Level Prediction Example

Invoice	Customer	Amount (\$000)	Due Date	Baseline Receipt	ML Predicted Receipt	Reason
AR-1001	Alpha Components	850	D+10	D+12	D+11	Usually pays 1-2 days late
AR-1002	Beta Retail	1,200	D+18	D+20	D+29	History shows ~11-day delay
AR-1003	Gamma Health	600	D+25	D+27	D+29	Slightly late strategic customer
AR-1004	Delta Tech	900	D+14	D+16	D+37	Delinquent / high-risk pattern
AR-1005	Epsilon Logistics	500	D+7	D+9	D+7	Auto-pay behavior

18. Lab Step 4 - Forecast Accuracy Comparison

Metric	Benchmark	ML Forecast	Improvement
Forecasted receipts D+1:D+45	\$9.3M	\$9.8M	+0.5M closer to expected actual
Actual receipts	\$9.9M	\$9.9M	-
Absolute error	\$0.6M	\$0.1M	\$0.5M lower
Accuracy	93.9%	99.0%	+5.1 pts
Largest miss	Delta Tech timing	Beta Retail timing	Reduced tail risk

Teaching point

A model should be adopted because it improves forecast outcomes, not because it is technologically sophisticated. Always compare ML to a transparent benchmark.

19. Lab Step 5 - Horizon Degradation Test

Horizon	ML Accuracy	Benchmark Accuracy	Recommendation
D+1:D+15	97.5%	91.0%	Use ML
D+16:D+30	95.8%	90.5%	Use ML
D+31:D+45	93.6%	89.8%	Use ML with review
D+46:D+60	86.2%	87.0%	Blend / consider other method

Because the dominant term is Net 45, model performance drops in the tail. A mature design may use ML in early periods, then switch to a driver/statistical method for later periods.

20. Scenario and Liquidity Analysis

Scenario	Receipt Timing Assumption	Closing Cash D+45	Liquidity View
Base ML	Use ML predicted dates	\$6.4M	Above buffer
Stress	Shift high-risk segment +7 days	\$5.1M	Tighter but acceptable
Management Action	Collections intervention on Delta/Beta	\$5.8M	Partial recovery
Benchmark	Average delay only	\$5.5M	Less differentiated customer behavior

Decision use

ML outputs should feed treasury decisions such as targeted collections actions, funding readiness, cash-buffer protection, and scenario-based liquidity management.

21. Method Blending Design

Forecast range	Preferred method	Reason
D+1:D+45	Short Term ML Prediction	Strong transaction and payment-behavior signal
D+46:D+60	Statistical / Driver blend	ML accuracy decay after dominant payment term
Beyond known transaction horizon	Driver / Trend / Statistical / Manual	Future cash increasingly depends on not-yet-created business events

- Blend methods by period range rather than forcing one method across the full horizon.
- Allow entity-specific method choices where business models differ.
- Retain a baseline version to measure forecast value add over time.

22. Model Governance and Monitoring

Control	Purpose
Training-data quality score	Detect missing lifecycle data and attribute drift.
Weekly retraining completion	Ensure model incorporates current trends and data drift.
Accuracy by horizon	Reveal where forecast utility degrades.
Accuracy by customer segment	Identify populations that require different treatment.
Exclusion-rule audit	Prevent atypical records contaminating training.
Benchmark comparison	Prove forecast value add.
Override tracking	Measure whether human adjustments improve or reduce accuracy.

23. Security, Roles, and Audit

Role	ML-related responsibility
Service Administrator	Environment configuration, forecast methods, integrations, job/security setup
Data / ERP Administrator	Source completeness, exclusion attributes, customer lifecycle integrity
Cash Manager	Review predictions, accuracy, exceptions and blended forecast
Controller	Approve scenarios and material overrides
Model Governance Owner	Validate performance, drift, fairness/segmentation and method suitability
Auditor	Review lineage, exclusions, retraining evidence and overrides

Audit evidence

Persist run date, training-window dates, eligible/excluded record counts, accuracy metrics, benchmark results, scenario/version, override reason, reviewer, and publication status.

24. Logs, Errors, and Troubleshooting

Log / Error Examples

ERROR ML-101 Training history below 18-month recommendation for entity NEW_GEO01.

WARN ML-118 Prediction horizon exceeds dominant customer pay term; expected accuracy may decline.

ERROR ML-204 Legacy-converted receivables were included in training despite exclusion rule.

INFO ML-001 Weekly model retraining completed; forecast-accuracy benchmark improved by 3.2 pts.

Symptom	Likely cause	Resolution
ML forecast unavailable	Feature not available in target release	Validate Oracle release capability; use alternate method until enabled
Low accuracy after D+45	Prediction horizon exceeds payment-term signal	Blend with another method for later periods
Sudden accuracy deterioration	Data drift or business-process change	Retrain, segment, investigate transaction/customer changes
New entity performs poorly	Insufficient comparable history	Build history or use benchmark/driver method
Immediate-pay segment causes late predictions	Mixed payment behaviors in training	Exclude or separately segment immediate-pay population
Legacy invoices distort model	Converted items included in training	Strengthen source/type exclusions and retrain

25. Production Runbook

1. Confirm ERP receivables lifecycle data is current and complete.
2. Run data-quality and exclusion controls.
3. Validate weekly model-training status and record counts.
4. Run or ingest ML predictions on the operational forecast cadence.
5. Compare ML accuracy with the approved benchmark and review horizon degradation.
6. Review customer/segment exceptions and liquidity impact.
7. Blend or override results where policy permits, with commentary and approval.
8. Publish the approved Rolling Forecast and archive run/model evidence.
9. After actual receipts are known, update forecast-accuracy scorecards and feed lessons into the next cycle.

26. Implementation Readiness Checklist

- ☐ Confirm actual ML feature availability for the target Oracle PCF release.
- ☐ Confirm Fusion Receivables business model is suitable for due-date-based ML.
- ☐ Obtain at least 18 months of clean historical transaction lifecycle data.
- ☐ Document customer segmentation and credit/collections profile quality.
- ☐ Configure and test exclusion rules for atypical transactions.
- ☐ Define training and prediction uptake cadence.
- ☐ Define benchmark methodology and accuracy thresholds.
- ☐ Design period-based blending for horizons where ML accuracy degrades.
- ☐ Build dashboard, audit, exception, and runbook controls.
- ☐ Complete UAT with Treasury, AR, Credit, Collections, Controller, and technical teams.

27. Advanced Student Challenge

Challenge

A new geography is moving onto Fusion Receivables. It has only six months of local history, but the parent company has 36 months of history from another geography. Most customers in the new geography use Net 30, while the mature geography uses Net 60.

- Decide whether you would train ML immediately, reuse the mature geography history, or use another method temporarily.
- Explain the risks of transferring behavior across geographies with different terms and customer mixes.
- Design a six-month ramp plan from Smart Driver / benchmark forecasting toward ML.
- Define the evidence required before declaring ML production-ready.

28. Summary and Completion Criteria

- The learner understands the documented short-term and mid-term ML forecasting concepts in PCF.
- The learner can evaluate ML eligibility from business-process and data-readiness perspectives.
- The learner can design training exclusions, retraining cadence, benchmark controls, and horizon blending.
- The learner can use ML forecast accuracy as a measurable decision criterion.
- The learner understands that current Oracle PCF documentation still contains a future-update caveat and will verify target-release availability before production use.
- The guide can function as product design documentation, implementation guide, training material, technical reference, and troubleshooting handbook.

End state

A production-ready ML forecasting design is not merely a model. It is a governed system of data quality, customer behavior, retraining, benchmarking, scenario analysis, liquidity decisions, and transparent fallback methods.